

Dot-env literacy

Toolchains need paths and flags, TOOLS, SIM, WAVE, without hard-coding secrets
in scripts you commit

Source, layers, and Git

- Source or dot loads the file into the current shell
- Commit a safe example template; keep the real env file gitignored
- Later layers win, local overrides, then shell exports
- Never commit tokens or private paths; document the keys teammates must set

Browser lab

Digital Design and Verification Platform

HomeTools.env

Learn load order, .env.example vs .env, and why secrets stay out of Git — resolve vars the way toolchains do.

Starter example: Clone project with .envr .example (safe), .envr (secrets), and optional .envr.local. Load env and watch which layer wins for each key.

Load starter example

Challenges 0 / 22 cleared

Quiz: template: Which file is the safe committed template? Answer: .envr .example

Answer:

Show hintCheckNextIdle

Quiz: template

Quiz: secrets file

Quiz: ignore

Quiz: order

First load

Spot the secret

Example not loaded

Local wins

Shell wins

Copy example

Quiz: syntax

Quiz: comment

Git safe

Red git add

Restore ignore

Write from env

Local wins

Quiz: commit example?

Exit B3

Disable local

Quiz: defaults

Full hygiene

Core ideas

Load order

Later wins: defaults → .env → .envr.local → shell exports.

Commit vs secret

Commit: .envr .example.Gitignore .envr / .envr.local.

Env files

.envr .example commit

.envr secrets

.envr.local envr

```
# local secrets - do NOT commit
tools/home/inf/inf
@Phylisville
secret
API_TOKEN=lab_secret_do_not_commit
```

cp .envr.example .env

Enable / write .envr.local

Disable .envr.local

Process & resolve

shell exports: (none)

not loaded - click Load / resolve env

export TOOLS=/opt/bin

unset shell exports

Load / resolve env

Toggle .gitignore for .envr

Simulate git add .envr

Resolved environment

Key	Value	Source
(load to see values)		

Git hygiene

.envr.example

tracked

(good)

.gitignore

has .env / .envr.local

.envr

ignored

(safe)

.envr.local

disabled

Log

```
$ state: .envr.example + .env with API_TOKEN;
.envr.local off
```

Cheat sheet

File	Meaning
EEETreeview	Assignment; no spaces around =
# comment	Ignored line
.envr .example	Safe template — commit it, not auto-loaded
.envr	Local secrets — gitignore
.envr.local	Machine overrides — later than .envr
Shell exports	Wins over file values in this tab

• Never put API tokens or passwords in committed files.

• If Make says TOOLS unset, you forgot to load [API] (or export).

• Example files document keys; real values live only on the machine.

Env file lab

learn_unix — env file lab

Real shell practice

```
wsl - module27-env-file-lab/examples/env_file

# Track A - real Unix
# real shell session (Track A)

$ pwd
/mnt/d/proj/designs/learning/courses/learn_unix/module27-env-file-lab/examples/env_file

$ ls -la
total 4
drwxrwxrwx 1 yongfu yongfu 4096 Jul 20 09:22 .
drwxrwxrwx 1 yongfu yongfu 4096 Jul 20 00:49 ..
-rwxrwxrwx 1 yongfu yongfu 1196 Jul 20 09:20 README.md
-rwxrwxrwx 1 yongfu yongfu 151 Jul 20 09:20 commands_to_try.txt
-rwxrwxrwx 1 yongfu yongfu 96 Jul 18 16:34 sample.env
-rwxrwxrwx 1 yongfu yongfu 369 Jul 18 16:34 use_env.sh

$ cat sample.env
# Sample env file for module8 env_file example
MY_VAR=hello
ANOTHER_VAR=world
# COMMENT=ignored

$ chmod u+x use_env.sh

$ ./use_env.sh
MY_VAR=hello
ANOTHER_VAR=world

$ source sample.env

$ echo MY_VAR=$MY_VAR
MY_VAR=hello

$ echo ANOTHER_VAR=$ANOTHER_VAR
ANOTHER_VAR=world
```

Real shell — source env

Real shell practice — try these

```
# pwd — print working directory (where am I?)  
pwd
```

```
# ls -la — list all entries, long format (what is here?)  
ls -la
```

```
# cat sample.env — show KEY=value lines (and comments)  
cat sample.env
```

```
# chmod u+x use_env.sh — make the helper script executable  
chmod u+x use_env.sh
```

```
# ./use_env.sh — source sample.env inside the script and print vars  
./use_env.sh
```

```
# source sample.env — load vars into the current shell
```

Pitfalls to watch

- Do not commit a real env file with secrets
- Do not put spaces around the equals sign in KEY equals value lines
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, clear a few challenges after the starter
- On the real shell, source the sample env and confirm the variables print
- When you are ready, take the short quiz, then continue to the course wrap